

2.1

$$B = \left(\begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 3 \\ 7 \end{pmatrix} \right) \quad u = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \quad v = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$$

$x_1, x_2, y_1, y_2 \in \mathbb{R}$

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podle tvrzení 8.50:

$$\langle u, v \rangle = [u]_B^* \cdot [v]_B$$

$$u = a \begin{pmatrix} 1 \\ 2 \end{pmatrix} + b \begin{pmatrix} 3 \\ 7 \end{pmatrix}$$

$$\left(\begin{array}{cc|c} 1 & 3 & x_1 \\ 2 & 7 & x_2 \end{array} \right) \sim \left(\begin{array}{cc|c} 1 & 3 & x_1 \\ 0 & 1 & x_2 - 2x_1 \end{array} \right)$$

$$b = x_2 - 2x_1$$

$$a = -3x_2 + 7x_1$$

$$[u]_B = \begin{pmatrix} 7x_1 - 3x_2 \\ -2x_1 + x_2 \end{pmatrix}$$

pro vektor v stejně akorát $x_1 \rightarrow y_1$ $x_2 \rightarrow y_2$ tedy: $[v]_B = \begin{pmatrix} 7y_1 - 3y_2 \\ -2y_1 + y_2 \end{pmatrix}$

$$\langle u, v \rangle = (7x_1 - 3x_2 \quad -2x_1 + x_2) \begin{pmatrix} 7y_1 - 3y_2 \\ -2y_1 + y_2 \end{pmatrix} =$$

$$= \underline{49x_1y_1} - \underline{21x_2y_1} - \underline{21x_1y_2} + \underline{9x_2y_2} + \underline{4x_1y_1} - \underline{2x_2y_1} - \underline{2x_1y_2} + \underline{x_2y_2} =$$

$$= \underline{\underline{53x_1y_1 - 23x_2y_1 - 23x_1y_2 + 10x_2y_2}}$$